

Deep Interior Differentiation of Water Worlds and its Impact on Planetary Radius

Fernanda Correa Horta¹ and Leslie Rogers¹

¹Department of Astronomy and Astrophysics, University of Chicago.

Abstract

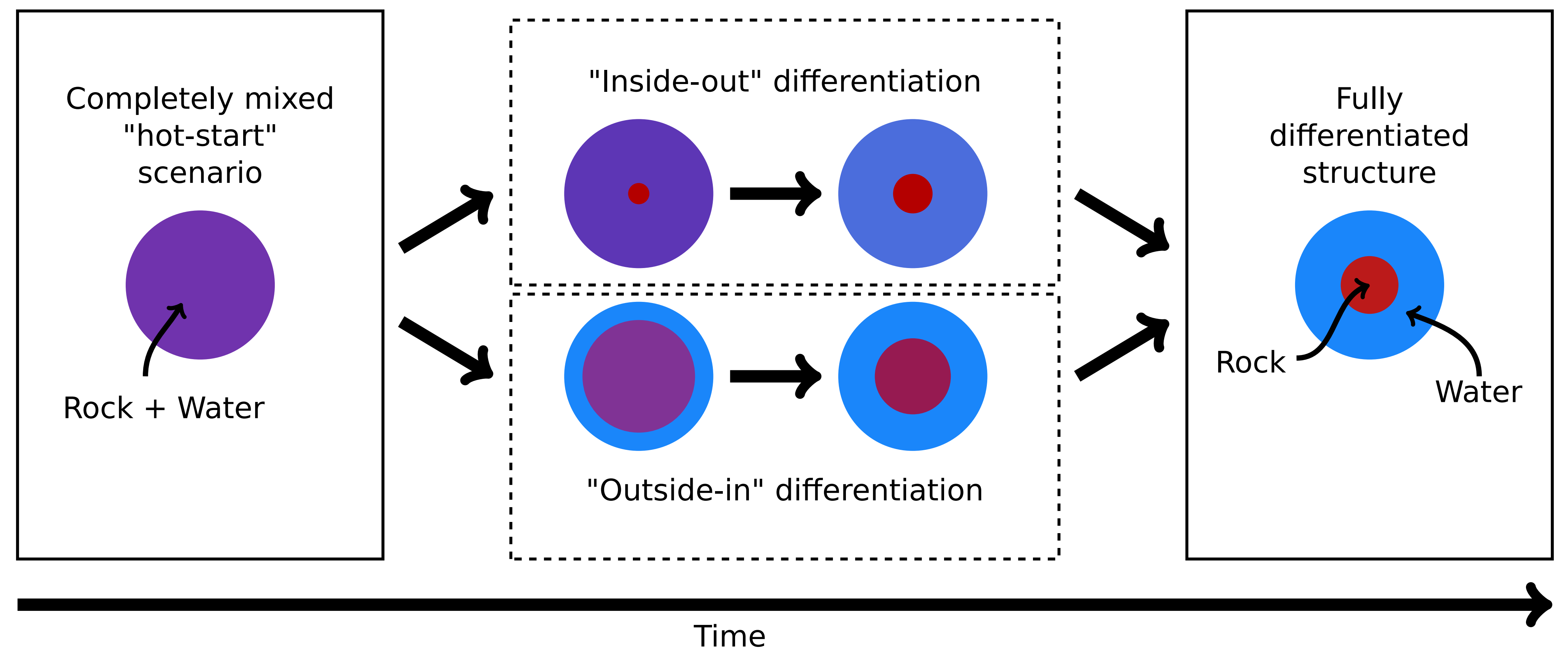
Characterizing and further understanding exoplanets' interiors and atmospheres is essential to advance our knowledge of planet formation and evolution. Traditionally, the interiors of water worlds are modeled with fully-differentiated layered structures. However, recent studies have shown that this strictly layered and differentiated structure might not accurately represent the interior of such planets. In this project, we relax the assumption of a completely differentiated structure and study the effects of a mixed-composition layer on the interior structure and evolution of water worlds. We find that, when considering an adiabatic temperature profile, an initially mixed planet would start differentiating from the inside-out, forming a rocky core with a water-enhanced mixed exterior as it cools down.

Motivation

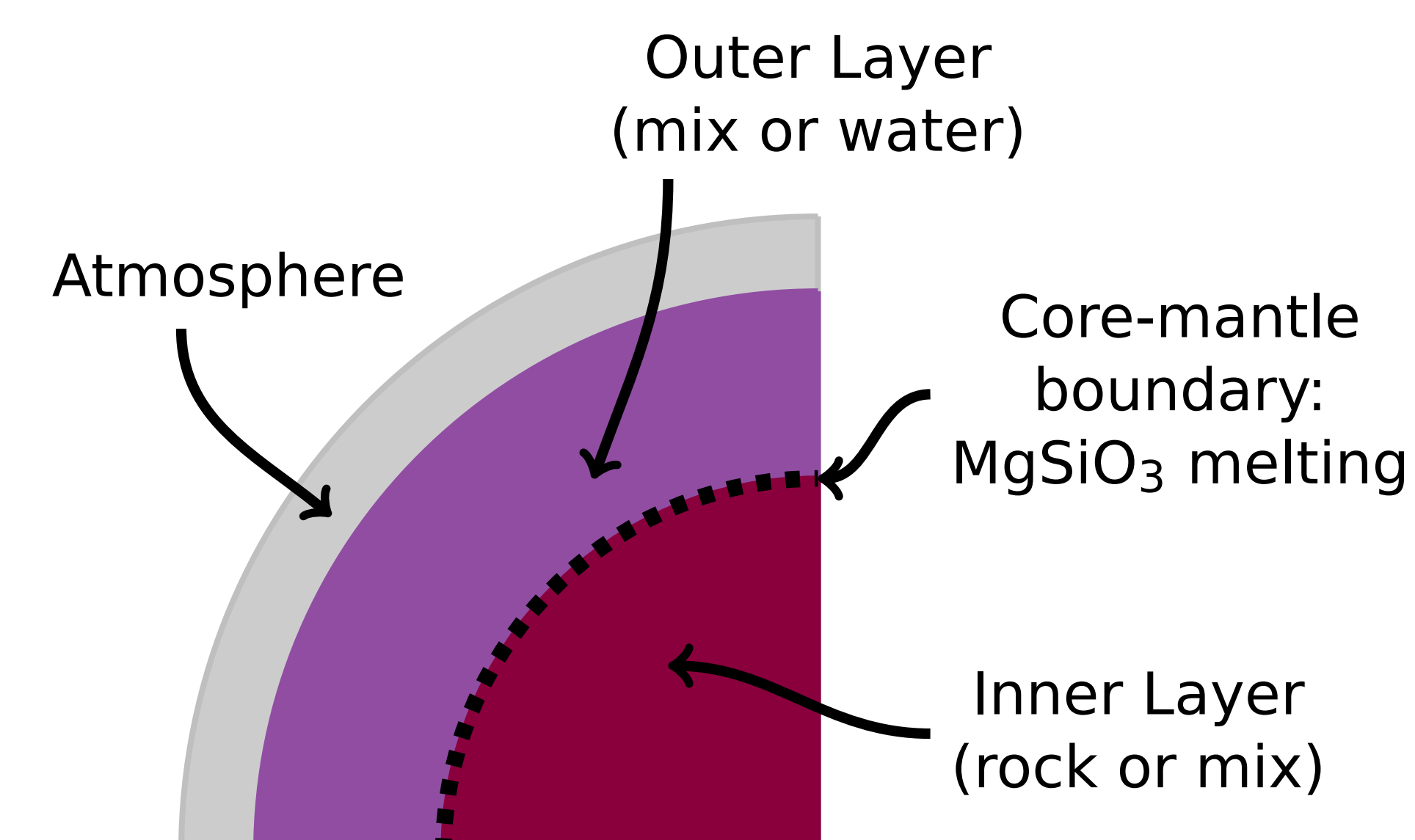
According to Kovačević et al. (2022), ice and rock are completely miscible for temperatures above the MgSiO_3 melting curve at high-pressure conditions (30 – 120 GPa). This result implies that water and rock may be mixed in the interiors of water worlds, challenging the strictly layered and differentiated structure frequently used to model planetary interiors.

Motivated by this, we propose a thought experiment where the deep interior of a planet, composed by only water (H_2O) and rock (MgSiO_3), starts completely mixed and differentiates into layers as it cools down.

Thought experiment



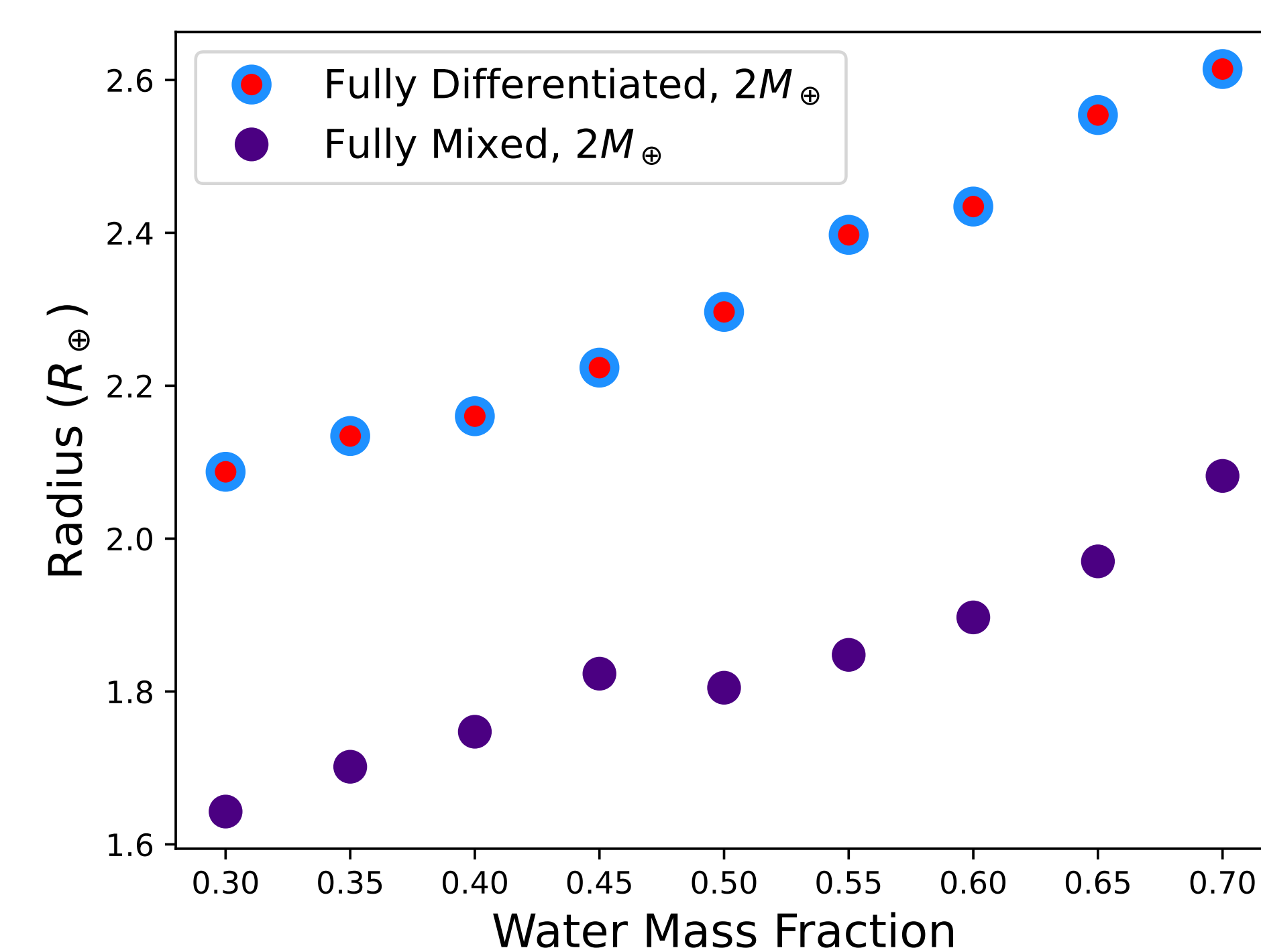
Model



- If the pressures and temperatures within a layer are favorable for miscibility, then that layer has a mixed composition.
- The interior of the planet follows an adiabatic temperature profile.
- We numerically solve the set of differential equations that model the interior structure of a planet, namely: hydrostatic equilibrium, mass continuity, heat transport and the appropriate equation of state (EoS) for each layer's composition.

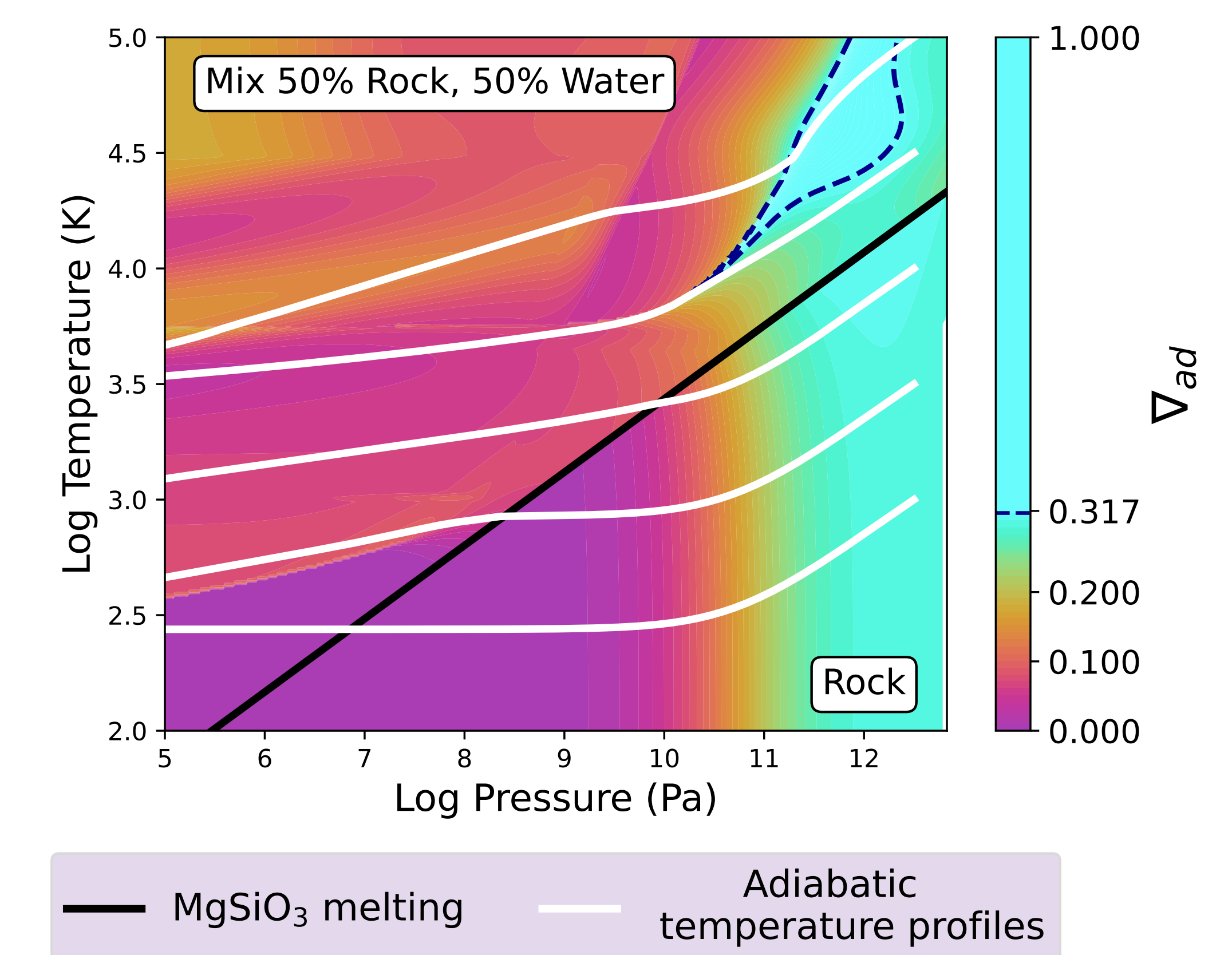
Radius Impact

We find that the deep interior of a planet, considering a fixed mass and overall composition (water mass fraction), has a different radius depending on the interior structure model. Particularly, a planet that is fully differentiated has a bigger radius than its fully mixed equivalent. We expect that the corresponding curve for planets with partially mixed structures to lay somewhere between the two end cases shown.



Adiabatic gradient maps

We construct tabulated EoSs for mixtures between H_2O and MgSiO_3 in different proportions using the additive volume law. For a pure H_2O layer, we use AQUA tables (Haldemann et al., 2020). For pure MgSiO_3 , we use Burnman (Cottaar et al., 2014) to obtain the properties of the material at any pressure-temperature condition.



In a log P-log T space, the MgSiO_3 melting curve has a slope of 0.317 (Fei et al., 2021). For all of our tables, the value of the adiabatic gradient of the water-rock mix, in most of the regions adjacent to the MgSiO_3 melting curve, is smaller than 0.317. Hence, any adiabatic temperature profile will have a slope shallower than the slope of the MgSiO_3 melting curve. This implies that all adiabatic interior temperature profiles that cross the MgSiO_3 melting curve, will do it from a high-pressure non-miscible zone (innermost layer of the planet) to a lower pressure zone where miscibility is favorable.

Conclusions

The inferred radius for a planet with a fixed mass and overall composition varies depending on the model we use for its interior structure. This motivates us to move on from the assumption of fully-differentiated planetary interiors and start including mixed composition layers into our models.

An initially fully-mixed planet with an adiabatic interior temperature profile will typically start differentiating from the inside-out as it cools down. In other words, the planet will form a rocky core surrounded by a mixed composition layer, which will become less rock abundant over time as the rock will settle to the core.

References

- Cottaar S., Heister T., Rose I., Unterborn C., 2014, *Geochemistry, Geophysics, Geosystems*, 15, 1164
 Fei Y., Seagle C. T., Townsend J. P., McCoy C. A., Boujibar A., Driscoll P., Shulenburger L., Furnish M. D., 2021, *Nature Communications*, 12, 876
 Haldemann J., Alibert Y., Mordasini C., Benz W., 2020, *A&A*, 643, A105
 Kovačević T., González-Cataldo F., Stewart S. T., Militzer B., 2022, *Scientific Reports*, 12, 13055

Contact

Fernanda Correa Horta
fpazcorrea@uchicago.edu

